Lab4: Clustering (and PCA)

STAT 4610/5610

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Emery Individual Section

```r
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.2
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
Attitude_clean <- read.csv("Attitude.clean.csv")
X <- dplyr::select(Attitude_clean, -Group)
X_scaled <- scale(X)

group_labels <- Attitude_clean$Group
```

Part 1. K-means Clustering

```r
set.seed(123)
km.out <- kmeans(X_scaled, centers = 3, nstart = 20)

# Compare clusters to known groups
table(Cluster = km.out$cluster, Group = group_labels)
```

```
       Group
Cluster IndoPost24 IndoPre25 IndoProf24 SLSp25 SLSp26 StatCollab
      1         23        28         17      8      8         25
      2          5        15          9      0      0          0
      3          2        12         10      3     13         12
```

Part 2. Comparison

```r
# Compare clusters to known groups
table(Cluster = km.out$cluster, Group = group_labels)
```

```
       Group
Cluster IndoPost24 IndoPre25 IndoProf24 SLSp25 SLSp26 StatCollab
      1         23        28         17      8      8         25
      2          5        15          9      0      0          0
      3          2        12         10      3     13         12
```

Using K-means clustering with K=3 and 20 random starts, the three clusters correspond somewhat with the 6 groups . Cluster 2 is the most distinct, containing only Indonesian respondents (5 IndoPost24, 15 IndoPre25, and 9 IndoProf24) and no American students. This suggests Indonesian respondents share a meaningfully different attitude profile compared to their American counterparts. Cluster 1 is the largest and most mixed, capturing some of every group, while Cluster 3 pulls in mostly American students (13 SLSp26 and 12 StatCollab) alongside some Indonesian students. Overall, K=3 reveals a meaningful distinction between Indonesian and American respondents, but a higher K would likely be needed to further differentiate within those two populations.

Part 3. Clustering Attitudes

```r
# Transpose so attitudes are the observations
X_t <- t(X_scaled)

set.seed(123)
km.att <- kmeans(X_t, centers = 3, nstart = 20)

# See which attitudes fall in which cluster
data.frame(Attitude = rownames(X_t),
           Cluster  = km.att$cluster) |>
  arrange(Cluster) |>
  split(~Cluster) |>
  print()
```

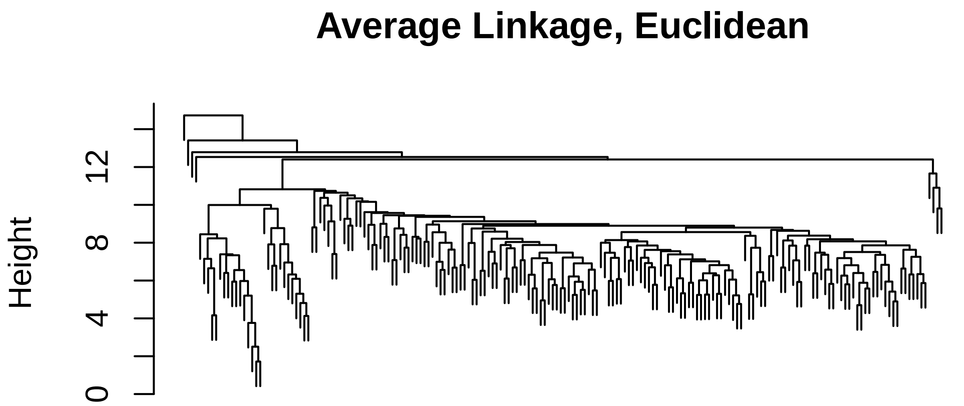
```
$`1`
          Attitude Cluster
A1.freq    A1.freq       1
A3.freq    A3.freq       1
A4.freq    A4.freq       1
A5.freq    A5.freq       1
A1.fav      A1.fav       1
A3.fav      A3.fav       1
A4.fav      A4.fav       1
A5.fav      A5.fav       1
A11.freq  A11.freq       1
A15.freq  A15.freq       1
A11.fav    A11.fav       1
A15.fav    A15.fav       1
A18.freq  A18.freq       1
A19.freq  A19.freq       1
A20.freq  A20.freq       1
A18.fav    A18.fav       1
A19.fav    A19.fav       1
A20.fav    A20.fav       1

$`2`
          Attitude Cluster
A2.freq    A2.freq       2
A2.fav      A2.fav       2
A8.freq    A8.freq       2
A9.freq    A9.freq       2
A10.freq  A10.freq       2
A8.fav      A8.fav       2
A9.fav      A9.fav       2
A10.fav    A10.fav       2
A22.freq  A22.freq       2
A24.freq  A24.freq       2
A22.fav    A22.fav       2
A24.fav    A24.fav       2

$`3`
          Attitude Cluster
A6.freq    A6.freq       3
A7.freq    A7.freq       3
A6.fav      A6.fav       3
A7.fav      A7.fav       3
A12.freq  A12.freq       3
```

```
A13.freq A13.freq       3
A14.freq A14.freq       3
A12.fav   A12.fav       3
A13.fav   A13.fav       3
A14.fav   A14.fav       3
A16.freq A16.freq       3
A17.freq A17.freq       3
A16.fav   A16.fav       3
A17.fav   A17.fav       3
A21.freq A21.freq       3
A23.freq A23.freq       3
A25.freq A25.freq       3
A21.fav   A21.fav       3
A23.fav   A23.fav       3
A25.fav   A25.fav       3
```

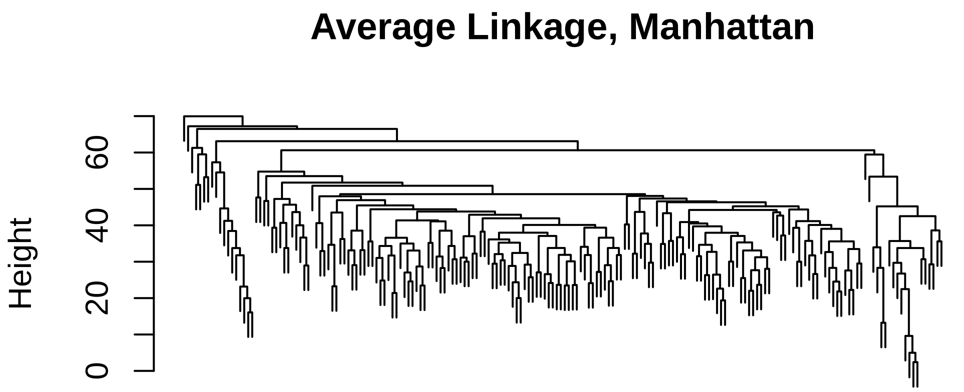
For every attitude, its freq and fav versions land in the same cluster. This suggests that how often respondents reported having an attitude is closely tied to how they believed it would impact collaboration. Cluster 1 contains attitudes A1, A3, A4, A5, A11, A15, A18, A19, and A20; Cluster 2 contains A2, A8, A9, A10, A22, and A24; and Cluster 3 contains A6, A7, A12-A14, A16, A17, and A21-A25. Without the actual attitude descriptions it is difficult to interpret what substantively distinguishes the three clusters, but the consistent freq/fav pairing suggests a strong relationship between frequency and perceived impact across all attitudes.

Part 4. Hierarchical Clustering - Average Linkage

```r
# Plot both dendrograms to visually choose K
# Euclidean distance
hc.avg.euc <- hclust(dist(X_scaled, method = "euclidean"), method = "average")

# Manhattan distance
hc.avg.man <- hclust(dist(X_scaled, method = "manhattan"), method = "average")


plot(hc.avg.euc, main = "Average Linkage, Euclidean", xlab = "", sub = "", labels = FALSE)
```




```r
plot(hc.avg.man, main = "Average Linkage, Manhattan", xlab = "", sub = "", labels = FALSE)
```




```r
best.k <- 3

cut.euc <- cutree(hc.avg.euc, k = best.k)
cut.man <- cutree(hc.avg.man, k = best.k)

table(Euclidean = cut.euc, Group = group_labels)
```

```
         Group
Euclidean IndoPost24 IndoPre25 IndoProf24 SLSp25 SLSp26 StatCollab
        1         30        55         34     11     21         37
        2          0         0          1      0      0          0
        3          0         0          1      0      0          0
```

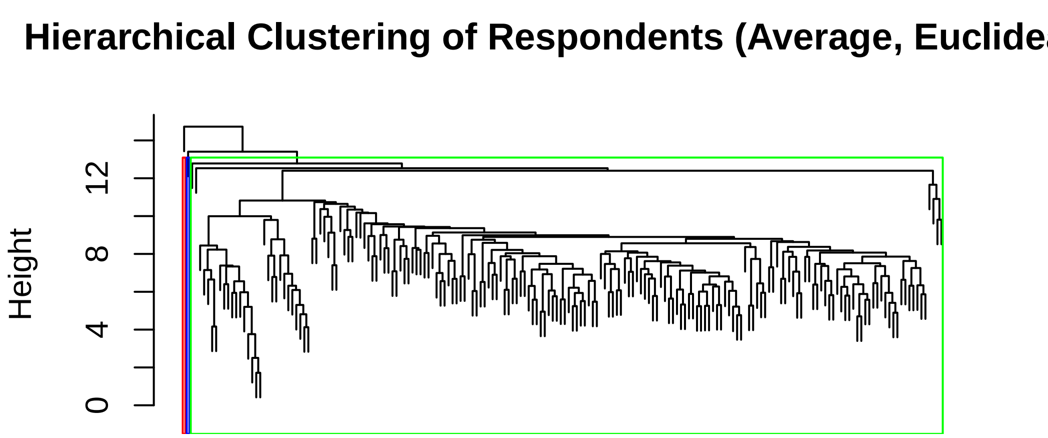
```r
table(Manhattan = cut.man, Group = group_labels)
```

```
         Group
Manhattan IndoPost24 IndoPre25 IndoProf24 SLSp25 SLSp26 StatCollab
        1         30        55         35     11     21         36
        2          0         0          0      0      0          1
        3          0         0          1      0      0          0
```

Both Euclidean and Manhattan distance gave identical results under average linkage, so Euclidean was kept as the default. K=3 was chosen to match the K-means analysis and allow for direct comparison, along wih the fact that it had one of the largest height gaos on the dendrogram. However, regardless of K chosen, average linkage placed nearly all respondents into one cluster with only a single outlier separated out. This is a known issue with average linkage. Complete linkage was tested and confirmed to give much more balanced clusters, so the poor result here is due to the linkage method rather than the data or distance measure.

Part 5. Best Dendrogram

```
plot(hc.avg.euc,
     main = "Hierarchical Clustering of Respondents (Average, Euclidean)",
     xlab = "", sub = "",
     labels = FALSE)

rect.hclust(hc.avg.euc, k = 3, border = c("red", "blue", "green"))
```



The dendrogram confirms what the table showed. Average linkage with K=3 places nearly all respondents into a single large cluster, with two individual IndoProf24 respondents split off as their own clusters. This does not correspond meaningfully to the six known groups at all. Compared to K-means, which at least identified a distinct Indonesian cluster, hierarchical clustering with average linkage fails to recover any meaningful group structure in this data. This reinforces that average linkage is a poor choice for this dataset, because average linkage tends to snowball in high-dimensional data, absorbing points one by one into one giant cluster rather than forming distinct groups.

Part 6. Cluster Attitudes

```
# Correlation-based distance
att.dist.corr <- as.dist(1 - cor(t(X_t)))
hc.att.comp.corr <- hclust(att.dist.corr, method = "complete")

# Euclidean for comparison
hc.att.comp.euc <- hclust(dist(X_t, method = "euclidean"), method = "complete")

# Cut at K=3
att.cut.corr <- cutree(hc.att.comp.corr, k = 3)
att.cut.euc  <- cutree(hc.att.comp.euc,  k = 3)

# Correlation distance
data.frame(Attitude = rownames(X_t), Cluster = att.cut.corr) |>
  arrange(Cluster) |>
  split(~Cluster) |>
  print()
```

```
$`1`
         Attitude Cluster
A1.freq   A1.freq       1
A3.freq   A3.freq       1
A4.freq   A4.freq       1
A1.fav     A1.fav       1
A3.fav     A3.fav       1
A4.fav     A4.fav       1
A8.freq   A8.freq       1
A8.fav     A8.fav       1
A11.freq A11.freq       1
A15.freq A15.freq       1
A11.fav   A11.fav       1
A15.fav   A15.fav       1
A18.freq A18.freq       1
A19.freq A19.freq       1
A20.freq A20.freq       1
A18.fav   A18.fav       1
A19.fav   A19.fav       1
```

```
A20.fav   A20.fav       1

$`2`
          Attitude Cluster
A2.freq   A2.freq       2
A5.freq   A5.freq       2
A2.fav     A2.fav       2
A5.fav     A5.fav       2
A9.freq   A9.freq       2
A10.freq A10.freq       2
A9.fav     A9.fav       2
A10.fav   A10.fav       2
A22.freq A22.freq       2
A24.freq A24.freq       2
A22.fav   A22.fav       2
A24.fav   A24.fav       2

$`3`
          Attitude Cluster
A6.freq   A6.freq       3
A7.freq   A7.freq       3
A6.fav     A6.fav       3
A7.fav     A7.fav       3
A12.freq A12.freq       3
A13.freq A13.freq       3
A14.freq A14.freq       3
A12.fav   A12.fav       3
A13.fav   A13.fav       3
A14.fav   A14.fav       3
A16.freq A16.freq       3
A17.freq A17.freq       3
A16.fav   A16.fav       3
A17.fav   A17.fav       3
A21.freq A21.freq       3
A23.freq A23.freq       3
A25.freq A25.freq       3
A21.fav   A21.fav       3
A23.fav   A23.fav       3
A25.fav   A25.fav       3
```

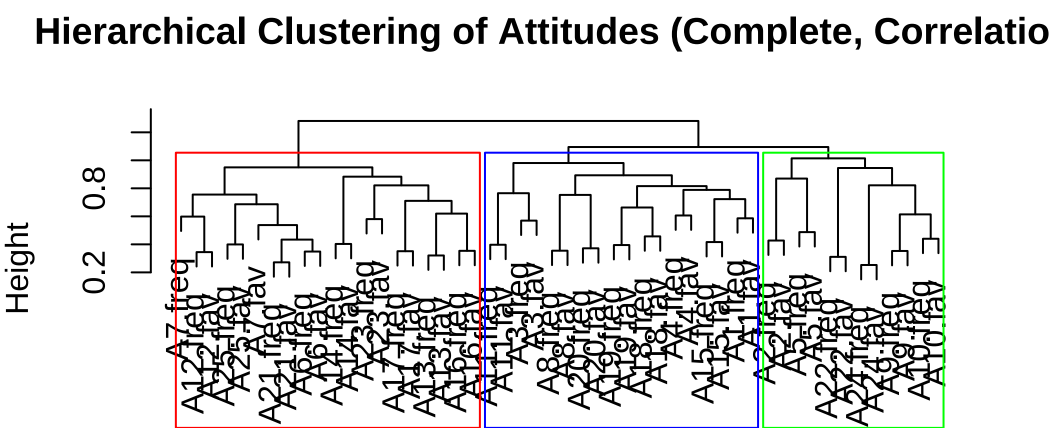
```r
# Count of attitudes per cluster
table(Cluster = att.cut.corr)
```

```
Cluster
 1  2  3
18 12 20
```

Correlation distance was used because we care about which attitudes move together across respondents, not their absolute values. Complete linkage was used instead of average linkage, which was shown earlier to chain badly on this dataset. The resulting three clusters are reasonably balanced with 18, 12, and 20 attitudes respectively. As with K-means, every attitude's freq and fav versions land in the same cluster, reinforcing that frequency and perceived impact are closely related.

Part 7. Best Dendrogram

```r
plot(hc.att.comp.corr,
     main = "Hierarchical Clustering of Attitudes (Complete, Correlation)",
     xlab = "", sub = "")

rect.hclust(hc.att.comp.corr, k = 3, border = c("red", "blue", "green"))
```



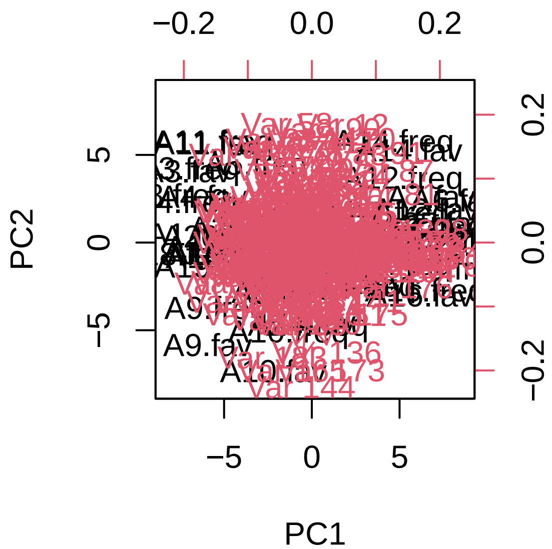
The dendrogram shows three decently balanced clusters with clear separation, suggesting complete linkage with correlation distance was a good fit for this data. The results closely mirror the K-means attitude clusters from Part 3 with the same attitudes tending to group together across both methods. This is strong evidence that these clusters reflect genuine structure in the data rather than an artifact of the algorithm chosen. As with K-means, every attitude's freq and fav versions land in the same cluster, reinforcing that frequency and perceived impact are consistently tied together regardless of the clustering method used.

Team Section Part 2: PCA

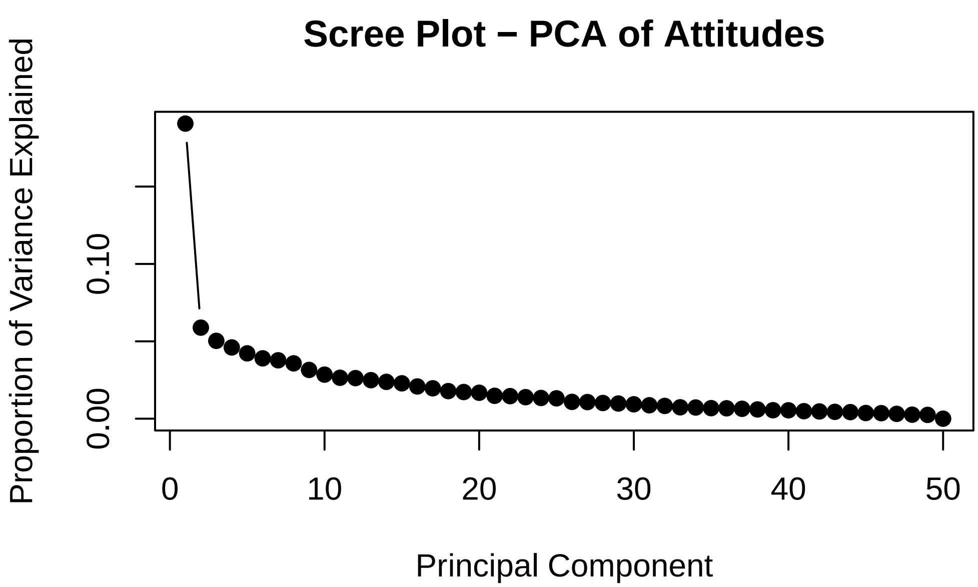
```r
# Run PCA on the attitudes (transposed matrix)
pca.att <- prcomp(t(X_scaled), scale. = FALSE)

# See how much variance each PC explains
summary(pca.att)
```

```
Importance of components:
                          PC1     PC2     PC3     PC4     PC5    PC6     PC7
Standard deviation     5.5654 3.09145 2.85836 2.73478 2.61845 2.5166 2.47599
Proportion of Variance 0.1907 0.05884 0.05031 0.04605 0.04222 0.0390 0.03775
Cumulative Proportion  0.1907 0.24955 0.29986 0.34591 0.38812 0.4271 0.46487
                           PC8     PC9    PC10    PC11    PC12    PC13   PC14
Standard deviation     2.41035 2.26225 2.15058 2.07276 2.06389 2.01153 1.9659
Proportion of Variance 0.03577 0.03151 0.02848 0.02645 0.02623 0.02491 0.0238
Cumulative Proportion  0.50064 0.53215 0.56063 0.58708 0.61331 0.63822 0.6620
                          PC15    PC16    PC17    PC18    PC19    PC20    PC21
Standard deviation     1.92631 1.83956 1.78652 1.70239 1.67289 1.65073 1.54973
Proportion of Variance 0.02285 0.02084 0.01965 0.01784 0.01723 0.01678 0.01479
Cumulative Proportion  0.68486 0.70570 0.72535 0.74320 0.76043 0.77720 0.79199
                          PC22    PC23    PC24    PC25    PC26    PC27    PC28
Standard deviation     1.53899 1.50297 1.47454 1.46297 1.32882 1.32151 1.28359
Proportion of Variance 0.01458 0.01391 0.01339 0.01318 0.01087 0.01075 0.01014
Cumulative Proportion  0.80658 0.82048 0.83387 0.84705 0.85792 0.86867 0.87882
                          PC29    PC30    PC31    PC32    PC33    PC34    PC35
Standard deviation     1.26218 1.22996 1.18717 1.15810 1.09413 1.08348 1.05338
Proportion of Variance 0.00981 0.00931 0.00868 0.00826 0.00737 0.00723 0.00683
Cumulative Proportion  0.88863 0.89794 0.90662 0.91488 0.92225 0.92948 0.93631
                          PC36   PC37    PC38    PC39    PC40    PC41    PC42
Standard deviation     1.04563 1.0194 0.98590 0.94442 0.93872 0.88927 0.87276
Proportion of Variance 0.00673 0.0064 0.00598 0.00549 0.00543 0.00487 0.00469
Cumulative Proportion  0.94304 0.9494 0.95542 0.96092 0.96634 0.97121 0.97590
                          PC43    PC44    PC45    PC46    PC47   PC48    PC49
Standard deviation     0.84639 0.83126 0.77342 0.76171 0.71350 0.6494 0.63031
Proportion of Variance 0.00441 0.00425 0.00368 0.00357 0.00313 0.0026 0.00245
Cumulative Proportion  0.98031 0.98457 0.98825 0.99182 0.99496 0.9976 1.00000
                            PC50
Standard deviation     2.472e-15
Proportion of Variance 0.000e+00
Cumulative Proportion  1.000e+00
```

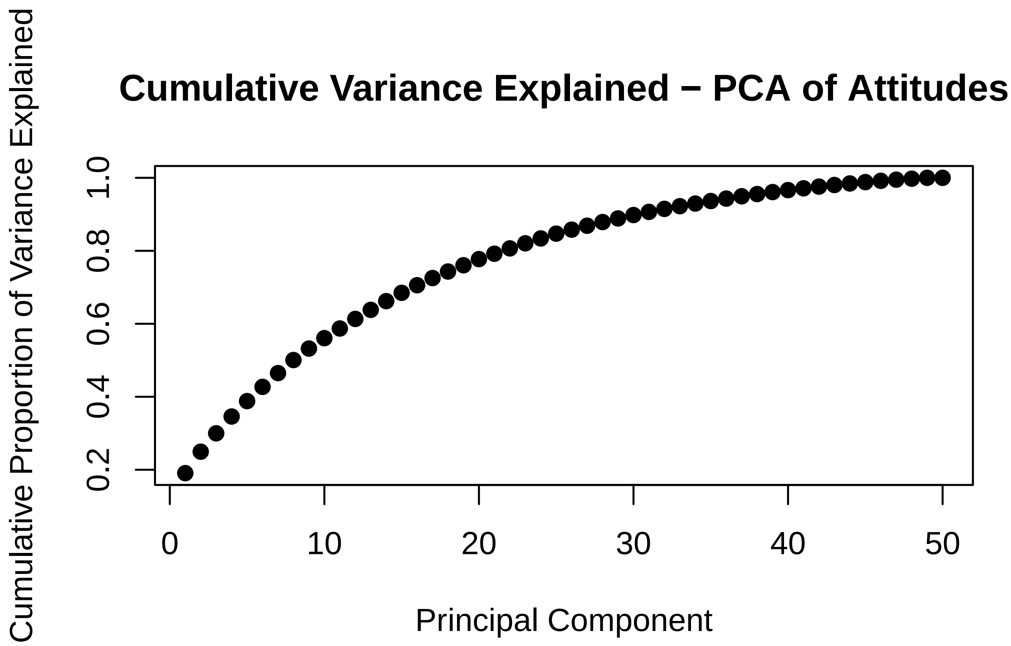
```r
# Plot the attitudes on PC1 vs PC2, labeled by attitude name
biplot(pca.att, scale = 0)
```



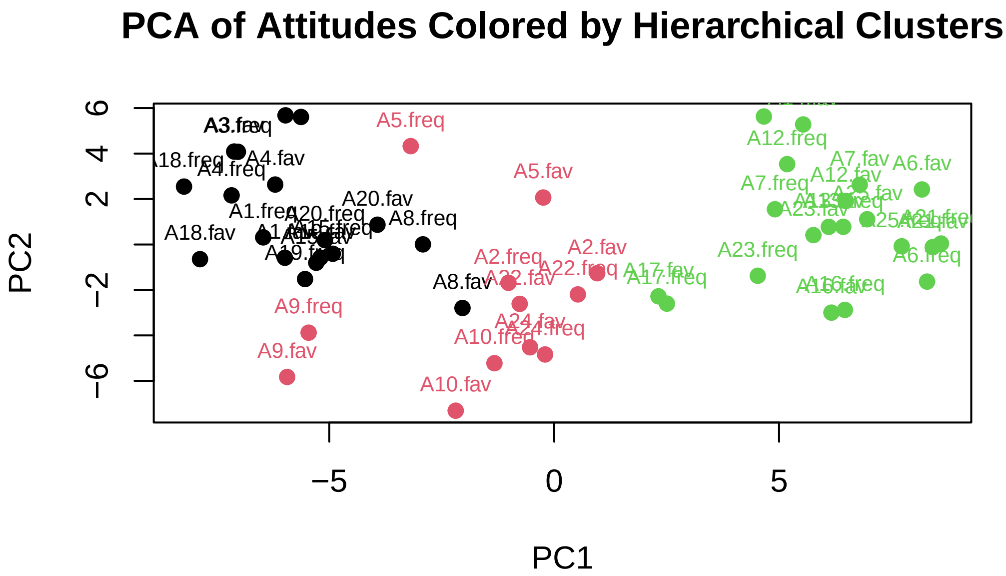

```r
# Scree plot
pve <- summary(pca.att)$importance[2,]  # proportion of variance explained
plot(pve, type = "b", pch = 19,
     xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     main = "Scree Plot - PCA of Attitudes")
```



```r
# Cumulative variance explained
plot(cumsum(pve), type = "b", pch = 19,
     xlab = "Principal Component",
     ylab = "Cumulative Proportion of Variance Explained",
     main = "Cumulative Variance Explained - PCA of Attitudes")
```




```r
# Color the PCA plot by clustering results to directly compare
plot(pca.att$x[, 1:2],
     col = att.cut.corr,
     pch = 19,
     xlab = "PC1", ylab = "PC2",
     main = "PCA of Attitudes Colored by Hierarchical Clusters")
text(pca.att$x[, 1:2],
     labels = rownames(pca.att$x),
     col = att.cut.corr,
     cex = 0.7, pos = 3)
```



The scree plot shows PC1 explains about 19% of the variance, and the cumulative variance plot shows you need over 20 components to explain 80% of the variance. This means the attitude data is spread across many dimensions and PCA alone does not give a simple, interpretable summary.

That being said, when the PCA plot is colored by the hierarchical cluster assignments, the three clusters separate cleanly in PC space with black on the left, pink in the middle/bottom, and green on the right. This is a good sign that the clusters reflect real structures in the data.

Overall, clustering is more interpretable than PCA here. The clusters give clear groupings of attitudes that are consistent across both K-means and hierarchical methods, while PCA requires too many components to be summarized easily. The PCA plot is most useful as a visual check that the clusters are genuinely distinct rather than as a standalone tool.